Week 6

Monday 2/16

Euler's method

$w_{i+1} = w_i + h\, f(t_i, w_i)$

$= w_i + h\lambda w_i$

$= (1+h\lambda) w_i$

$\rightarrow w_1 = (1+h\lambda) w_0$

$w_2 = (1+\lambda h)^2 w_0$

$w_3 = (1+h\lambda)^3 w_0$

$w_{i+1} = (1+\lambda h)^{i+1} w_0$

Geo Sequence

if we want $w_{i+1} \to 0$ this is what happens when h is what values?

st $|1+\underbrace{h\lambda}| < 1$

def the stability domain: values of $h\lambda \in \mathbb{C}$ for which $w_i \to 0$ $i \to \infty$ (correct asymptotic behavior)

Trapezoid Method (im)

$w_{i+1} = w_i + \frac{h}{2}(f_i + f_{i+1})$

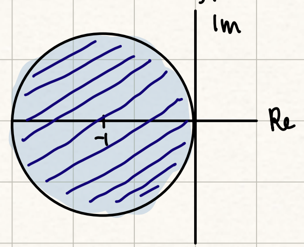
Wednesday 2/18

Stability Domain of a numerical ODE solver is the set of all $\lambda, h \in \mathbb{C}$ s.t. $\lim_{i \to \infty} w_i = 0$

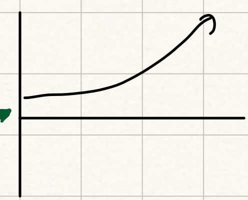
correction: $w_{i+1} = w_i + h \underbrace{f_i}_{\lambda w_i}$

$|1 + \lambda h| < 1$ $\quad z = \lambda h$

$|z - (-1)| < 1$



sufficient to consider $y' = \lambda y$ $\lambda < 0$ since otherwise



ex) implicit trap. method ← a way to relax "strict" constraints on h

$w_{i+1} = w_i + \frac{h}{2}(f_i + f_{i+1})$

→ what is the stability domain

let $f_i = \lambda y_i$

$w_{i+1} = w_i + \frac{h}{2}(\lambda w_i + \lambda w_{i+1})$

$w_{i+1} = w_i\left(1 + \frac{h\lambda}{2}\right) + \frac{h\lambda}{2} w_{i+1}$

$w_{i+1}\left(1 - \frac{h\lambda}{2}\right) = \left(1 + \frac{h\lambda}{2}\right) w_i$

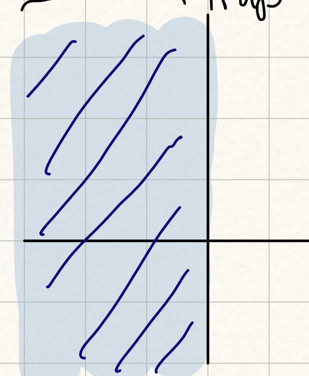
$w_{i+1} = \underbrace{\left(1 + \frac{h\lambda}{2}\right) / \left(1 - \frac{h\lambda}{2}\right)}_{r \text{ in geometric sequence}} w_i$

$\left.\begin{array}{l} \lambda < 0 \\ \uparrow \text{ real part } < 0 \end{array}\right.$

$w_{i+1} = \left(\frac{1 + \frac{h\lambda}{2}}{1 - h\lambda/2}\right)^{i+1} w_0 \qquad |r| < 1 \quad \therefore$

$\left|\frac{1 + \frac{h\lambda}{2}}{1 - h\lambda/2}\right| < 0$ $\quad$ since $\lambda < 0$, this is easy b/c $1 - \frac{h\lambda}{2} > 1 + \frac{h\lambda}{2}$

stability domain of implicit trap



$\therefore$ all values of λh (assuming $\lambda < 0$) will go to 0

$-1 < \frac{1 + h\lambda/2}{1 - h\lambda/2} < 1$

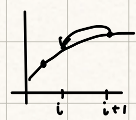
$\frac{h\lambda}{2} - 1 < 1 + h\lambda/2 < 1 - \frac{h\lambda}{2}$ $\qquad$ subtract $\frac{\lambda h}{2}$

$\cancel{-1} < 1 < 1 - h\lambda$

$0 < -\lambda h$ $\quad$ since $\lambda < 0$ & $h > 0$, then $-\lambda h > 0$ no matter what $\square$

Ex) Backward Euler:

$w_{i+1} = w_i + hf_{i+1}$ (becomes implicit)



$w_{i+1} = w_i + h\lambda w_{i+1}$

$(w_{i+1})(1-\lambda h) = w_i$

$w_{i+1} = \frac{1}{1-\lambda h} w_i$

$w_{i+1} = \left(\frac{1}{1-\lambda h}\right)^{i+1} w_0$

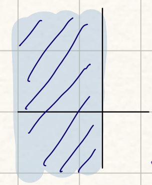
$\to -1 < \frac{1}{1-\lambda h} < 1$

$\lambda h - 1 < 1 < 1 - \lambda h$

$-1 < 1 < 1 - 2\lambda h$

$0 < -2\lambda h$

$\to$ since $\lambda < 0$ & $h > 0$, always true



stability domain of λh ☺

In order to deal w/ stiff ODEs...

1) truncation error $\frac{h^n}{n!} y^n$

2) constraints on $h\lambda \in \mathbb{C}$

how to apply implicit methods

how to solve for w_{i+1}:

ex) Backward Euler

$w_{i+1} = w_i + hf_{i+1}$

friday 2/20

$w_{i+1} = w_i + hf_{i+1}$

→ how to solve when $f(t,y)$ is nonlinear

cast as a root-finding problem

$F(w_{i+1}) \equiv w_{i+1} - w_i - hf(t_{i+1}, w_{i+1}) = 0$

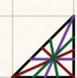
→ solve for w_{i+1} by defining $g(w_{i+1})$ such that $g(w_{i+1}) = w_{i+1} \Rightarrow g(w) = w$

→ $\underbrace{g \in C[a,b]}_{\text{continuous}}$, $\underbrace{g \in C'[a,b]}_{\text{differentiable}}$, $\underbrace{g(w) \in [a,b]}_{\text{belonging to same int}}$, $\underbrace{|g'(w)| < 1 \quad \forall w \in (a,b)}_{\text{bounded derivative}}$

→ implies that a unique soln exists to $g(w) = w$

$\lim_{k\to\infty} g(w_k) = w$

→ Use Newton's Method: $g(w) \equiv w - \frac{F(w)}{F'(w)}$



$$w_{i+1}^{k+1} = w_{i+1}^{k} - \left[\frac{w_{i+1}^{k} - w_i - hf(t_{i+1}, w_{i+1}^{k})}{1 - h(f_y(t_{i+1}, w_{i+1}^{k}))}\right]$$

... how do we know when to stop...

$$\left| w_{i+1}^{k} - w_{i+1}^{k+1} \right| < \text{tolerence}$$

usually only needs a few iterations → start w_1 w_i

How to do Imp. Trap.

$$w_{i+1} = w_i + \frac{h}{2}(f_i + f_{i+1})$$

$$F(w_{i+1}) \equiv w_{i+1} - w_i - \frac{h}{2}(f_i + f_{i+1}) = 0$$

$$g(w) = w - \frac{F(w)}{F'(w)}$$

$$w_{i+1}^{k+1} = w_{i+1}^{k} - \frac{w_{i+1}^{k} - w_i - \frac{h}{2}(f(t_i, w_i) + f(t_{i+1}, w_{i+1}^{k}))}{1 - \frac{h}{2}(f_y(t_{i+1}, w_{i+1}^{k}))}$$